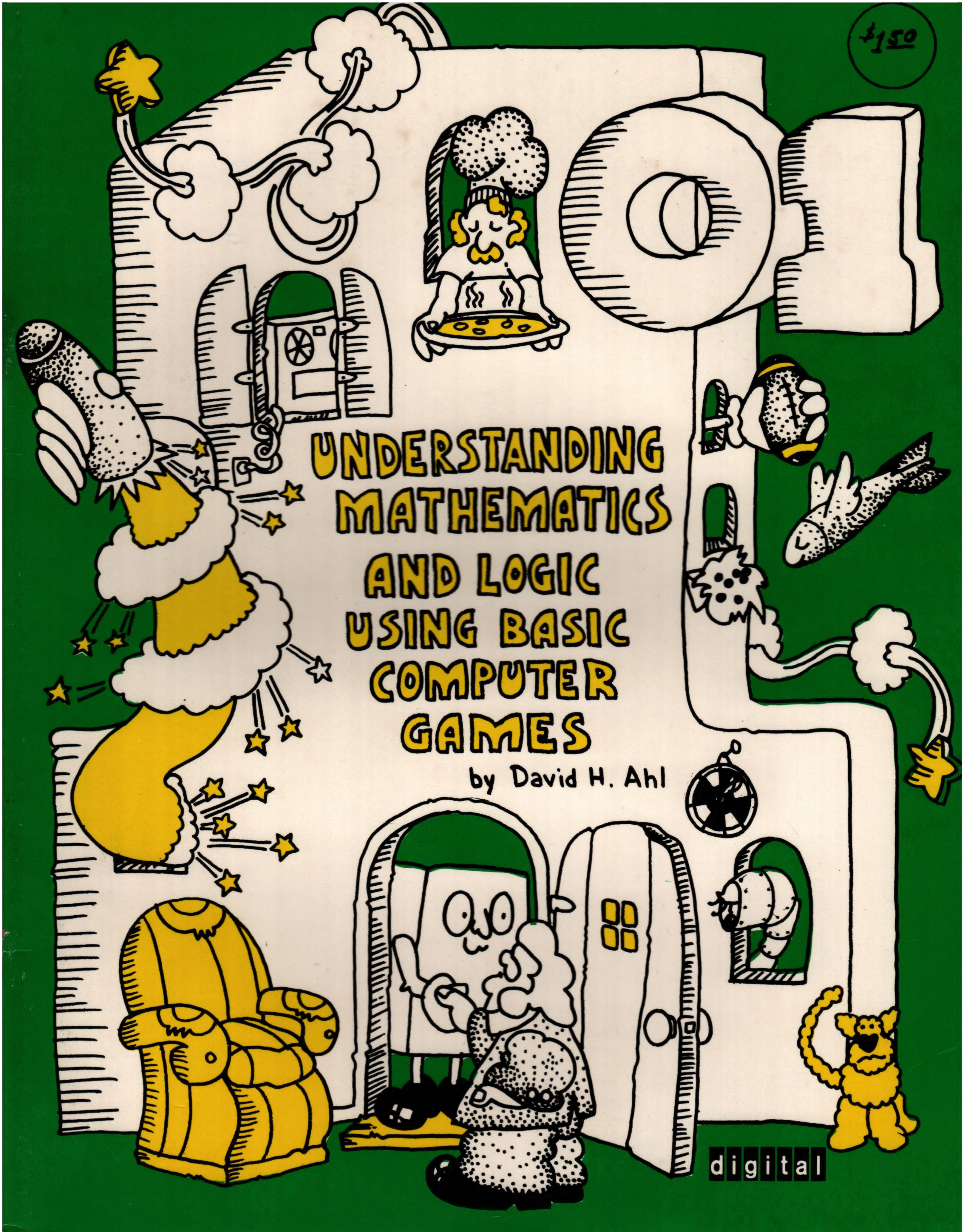
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UNDERSTANDING MATHEMATICS AND LOGIC USING BASIC COMPUTER GAMES
by David H. Ahl
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by

David H. Ahl

Digital Equipment Corporation
Maynard, Massachusetts

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Preface

My original intent with this book was to write a brief 12-page or so supplemental teacher's guide to go along with 101 BASIC Computer Games. However, one idea led to another and indeed the only way this book stopped at 60 pages was to confine it to mathematics and logic and to regard it only as a jumping off point in using games in learning.

Understanding Mathematics and Logic Using BASIC Computer Games is not only a teachers guide; it is a student workbook and background resource manual as well. It has lots of exercises to make learning fun. It has lots of supplemental projects to keep curious minds active. And it has lots of ideas that should lead to other worthwile adventures in learning and understanding.

In Future Shock, Alvin Toffler said, "Tomorrow's schools must teach not only data, but ways to manipulate it. Students must learn how to discard old ideas, and how and when to replace them. They must, in short, learn how to learn."

Games help people learn to learn. As you'll see from the introduction, even traditional learning is enhanced and improved by the use of games. But beyond that, the idea of this book is to better prepare people for the future by helping them to think, to be adaptable, to expect surprises, to understand, and to have fun.

Maynard, Massachusetts
November, 1973

David H. Ahl

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The Game of Life (discussion):

Paul Shapiro
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Projectile Motion:

<u>University Physics</u>
by Sears and Zemansky

Battle of Numbers, Nim:

<u>BASIC Programming, Second Edition</u>
by Kemeny and Kurtz

Contents

Introduction

History of Sports and Games

Sports and games have been with us almost since the dawn of man. The ancient Chinese and Egyptians devised an astonishing array of mathematical and logical games (Nim, Towers of Hanoi, Awari or Kalah, etc.). The Greeks originated a great many physical games in their Olympic competitions. Medieval Europe was responsible for many games played for recreation in nobles' courts (Chess, Hi-Q, etc.). Team games are a phenomena of the last few centuries with many originating in England and the United States.

Throughout history, the common thread running through all sports and games is that their intended function has been purely recreational. Games were viewed as a diversion from the realities of life. Secondly, sports and games were and are a leisure-time activity. In early Europe, only the nobles, who had leisure time, played games. Even today, except for professional athletes, sports and games are generally considered outside, extra-curricular activities.

Nevertheless, sports and games have generally had some redeeming virtues. Merely serving as a break from the realities of life is probably enough to assure games a place in history. But there is more. Chinese philosophers spoke of games as sharpening one's wit. Sports build the body just as card, board, and mathematical games build the mind. Team games heighten the spirit of trust and cooperation.

Games as an Educational Tool

Not until the last 10 or 15 years of educational innovation, however, have games ever been used primarily as an educational tool where learning is the primary purpose of playing. This has now happened and there is a growing body of evidence that indicates that a combination of learning games and student teams may well be one of the most effective approaches to learning ever devised. In a study by Edwards, DeVries, and Snyder (1972), the games-teams combination in seventh grade classes resulted in a significant increase on several dimensions of mathematics achievement. A follow-up study in 1973 found that both games and teams represent useful techniques for restructuring classroom processes. Their effects are complementary in that they create very different classroom experiences for students. Games cause students to translate their increased interpersonal interaction into increased informal peer-tutoring on the subject material at hand. They are also likely to view their class in a much more positive light. The addition of teams to a traditional classroom meets a rather different set of objectives. Attending the class and encountering the subject matter is not necessarily made more fun as it is with games. However, students work together on traditional classroom tasks to a much greater degree resulting in an increased level of mutual concern among the students.

Other studies echo the results above. The learning effectiveness of games have been frequently cited (Allen, Allen, and Ross, 1970; Boochock and Schild, 1968; Fletcher, 1971). Learning games generally create an intense and often enjoyable interpersonal experience. This is due in part to the interdependent task structure which requires interaction among the players (Inbar, 1968).

Teams have been in use longer as an educational technique. Generally it has been found that students in a team outperform students working as individuals. A key reason for the effect of teams has been the high rate of peer tutoring (Wodarski, Hamblin, Buckholdt, and Ferritor, 1971).

As mentioned earlier, because games and teams positively affect different classroom processes, combining the two techniques creates an even more powerful effect on the learning environment. This in turn is translated into greater academic growth as well as increased trust and cooperation.

Some Definitions

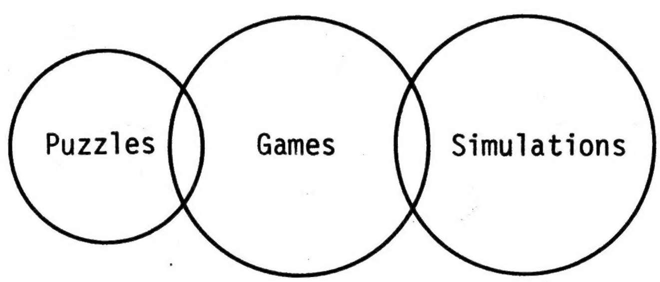
In this book a number of terms are used which are sometimes confused with one another.

A Puzzle is a problem that has a baffling quality or great intricacy, that one must exercise substantial mental ingenuity and thought to solve.

A Game is an amusement or sport involving competition under a specific set of rules. The competition may be against oneself or against laws of chance such as in solitaire or roulette.

A Simulation is a model or representation of a real-life process, situation, or event, frequently on a computer.

One might think of puzzles, games and simulations as having a certain overlap as follows:



This book deals principally with games, mostly on the computer. Some computer games are simulations of their real-life counterpart (Football, Checkers, Poker, etc.). Other computer games are simulations of other real processes with a recreational quality (Lunar Landing, Parachute Jump, Artillery Weapon Firing, etc.). Another group of games are actually puzzles (Caves, 1 Check, Guess, etc.). While still another group of games are devised solely for their recreational and mind-broadening value (Game of Life, Bagels, Bulls and Cows, etc.)

Ideas and Resources

For the most part the computer is finding its way into schools as a supplemental tool. Rare is the case where the curriculum is planned around a computer resource. Consequently, rather than try to mold a curriculum to a computer resource or vice versa for that matter, this book presents a verbal supermarket of ideas which can be integrated into curricula wherever seems appropriate. Ideas are suitable over a wide range of subject areas and grade levels, from 7th grade math to 12th grade social studies.

Conversely, the book can serve as the basis for an elective computer literacy course. In this manner it should probably be used after students have learned the rudiments of programming in BASIC.

Understanding Mathematics and Logic Using BASIC Computer Games is a companion volume to 101 BASIC Computer Games. Listings and sample runs of the games are contained in the latter volume; hence, it is highly recommended that both volumes be used.

Guessing Games - An Introduction to Mathematical Logic

Telephone Books

Let's say we wanted to look up the phone number of John Maddock in the phone book. First we would open the book about half way through because we know that M is about in the middle of the alphabet. Or if we wanted Mary Garvin's number we would open the book about a quarter of the way because G is about 1/4 through the alphabet. This is a reasonable strategy to follow when we know what we're looking for and roughly the structure of the list.

Now let's say we want to rent a car for a week because ours just got racked up in an accident. We would probably go to the Yellow Pages. Since we happen to have a National Car Rental credit card, we'll call them first. Since N is about 1/2 way through the alphabet we should thumb about halfway through the category "Automobile Renting and Leasing," right? Wrong. In the Philadelphia Yellow Pages, National is on the 14th page out of 17 in the category. Why? Two reasons:

1. There are loads of companies who deliberately start their name with A so they'll be at the beginning of the category where you're likely to look first if you had no company already in mind (names like AA Auto Leasing, A-D Rental, Ace, Adam Leasing, Alray, etc.)

2. The large display ads are generally clustered at the beginning of the category.

So our search technique must be different when the list is structured differently.

Consider another example. Suppose one summer we got a job at the local telephone company business office. There, they have books called reverse directories. There are two types: one lists phone numbers and names by address. The other lists names and addresses by phone number. Suppose someone calls the operator to report a fire and is so panicky that they don't give their name and address clearly, but the operator does manage to get their phone number. You have to look up this number in the reverse directory in a hurry and get the name and address to report to the fire department. The number is 561-7845. You quickly reason that phone numbers can be anything between 111-1111 and 999-9999 and reason that 561 numbers should be about 1/2 way through. Opening the book half way, you find 232 numbers. What happened? Obviously, the list is structured differently than you thought. After you finally find the number, you decide to see what went wrong.

Here are the phone exchanges in your town:

172	234
173	235
175	237
191	382
192	383
194	384
211	512
212	513
230	561
231	562
232	931

So you can see that 561 is way at the end of the list. Searching for a social security number or any other in which the list is unstructured will lead to similar problems.

Being a smart person who knows that computers ought to be able to do jobs like looking up numbers in a hurry you decide that when you get back to your school computer in the fall, you're going to find out more about searching for numbers in a list.

I'm Thinking of a Number

Before we can design a program to search for numbers in a list, we must have a good search technique. Interestingly enough it turns out that search techniques for finding a known number in a list of unknown structure are similar to the techniques for finding an unknown number in a list with known structure. As you get further into this unit, you'll see why this is true. For now, let's think about finding an unknown, mystery munber in the list 1 to 100.

EXERCISE - 1

Divide the class into small groups or teams of 2 or 3 members each. Play the computer program GUESS. This program asks you to find a mystery number between 1 and 100. After each guess, the computer tells you if you are too high or too low. Play 10 games. How many guesses does it take on average to find the number? Can you tell why it shouldn't take more than 7 guesses to get the number?

EXERCISE - 2

Change the limit in the program to guess a number between 1 and 1000. If it took 7 guesses to get the number between 1 and 100, should it take 7 x 10 = 70 guesses to get this one? Hardly. But why not? While there are many search techniques, one is clearly best in this case. If you understand how the method works, you should be able to describe why you will never need more than 10 guesses to find any number between 1 and 1000.

```
RUNNH
I HAVE CHOSEN AN INTEGER 0 THROUGH 100.   TRY TO
GUESS MY NUMBER IN AS FEW TRIES AS POSSIBLE.

YOUR GUESS IS? 10
TOO HIGH
YOUR GUESS IS? 1
TOO LOW
YOUR GUESS IS? 2
TOO LOW
YOUR GUESS IS? 3
TOO LOW
YOUR GUESS IS? 4
CORRECT IN 5 GUESSES.
GOOD JOB.

LET'S PLAY AGAIN
I'M THINKING OF ANOTHER NUMBER
YOUR GUESS IS? 10
TOO LOW
YOUR GUESS IS? 20
TOO LOW
YOUR GUESS IS? 30
TOO LOW
YOUR GUESS IS? 40
TOO LOW
YOUR GUESS IS? 50
TOO LOW
YOUR GUESS IS? 60
TOO LOW
YOUR GUESS IS? 70
TOO LOW
YOUR GUESS IS? 80
TOO HIGH
YOUR GUESS IS? 75
TOO HIGH
YOUR GUESS IS? 73
TOO HIGH
YOUR GUESS IS? 72
TOO HIGH
YOUR GUESS IS? 71
CORRECT IN 12 GUESSES.
BUT YOU SHOULDN'T NEED MORE THAN 7 GUESSES.

LET'S PLAY AGAIN
I'M THINKING OF ANOTHER NUMBER
YOUR GUESS IS? 50
TOO HIGH
YOUR GUESS IS? 25
TOO LOW
YOUR GUESS IS? 37
TOO HIGH
YOUR GUESS IS? 32
TOO HIGH
YOUR GUESS IS? 29
CORRECT IN 5 GUESSES.
GOOD JOB.
```

I'm Thinking of a Letter

All lists are not made up of numbers even though it may seem like it with bank accounts, charge accounts, telephone numbers, license plates, social security and all kinds of other things that assign numbers to people. Nevertheless, we still describe the vast majority of the world by name (people, things, animals, etc.). As we saw with a phone book, we should be able to search for letters (or words) like numbers. Let's see.

EXERCISE - 3

In the same or different teams, play the game LETTER on the computer. How many guesses at most should it take to find a mystery letter out of the English 26-letter alphabet? Would it change if we had the 31-letter Norwegian alphabet? How about for the 33-letter Russian alphabet?

```
LETTER GUESSING GAME

I'LL THINK OF A LETTER OF THE ALPHABET, A TO Z.
TRY TO GUESS MY LETTER AND I'LL GIVE YOU CLUES
AS TO HOW CLOSE YOU'RE GETTING TO MY LETTER.

OK, I HAVE A LETTER.  START GUESSING.

WHAT IS YOUR GUESS? Q

TOO HIGH.  TRY A LOWER LETTER.

WHAT IS YOUR GUESS? H

TOO HIGH.  TRY A LOWER LETTER.

WHAT IS YOUR GUESS? F

TOO HIGH.  TRY A LOWER LETTER.

WHAT IS YOUR GUESS? A

TOO LOW.  TRY A HIGHER LETTER.

WHAT IS YOUR GUESS? D

TOO HIGH.  TRY A LOWER LETTER.

WHAT IS YOUR GUESS? B

TOO LOW.  TRY A HIGHER LETTER.

WHAT IS YOUR GUESS? C


YOU GOT IT IN 7 GUESSES!!
BUT IT SHOULDN'T TAKE MORE THAN 5 GUESSES!
```

Trap a Number

There is no reason to limit ourselves to making one guess on each turn of GUESS assuming we structure the program differently. The program TRAP does this, i.e., it permits two guesses per turn and then tells where the unknown number is relative to the two guesses.

EXERCISE - 4

With the same teams, play the game of TRAP on the computer. You should need fewer turns to find the mystery number than with GUESS. Why? Several strategies for playing TRAP work fairly well, however, one is clearly the best. Can you determine what it is?

```
WANT INSTRUCTIONS (1 FOR YES)? 1
I AM THINKING OF A NUMBER BETWEEN 1 AND 100
TRY TO GUESS MY NUMBER. ON EACH GUESS,
YOU ARE TO ENTER 2 NUMBERS, TRYING TO TRAP
MY NUMBER BETWEEN THE TWO NUMBERS. I WILL
TELL YOU IF YOU HAVE TRAPPED MY NUMBER, IF MY
NUMBER IS LARGER THAN YOUR TWO NUMBERS, OR IF
MY NUMBER IS SMALLER THAN YOUR TWO NUMBERS.
IF YOU WANT TO GUESS ONE SINGLE NUMBER, TYPE
YOUR GUESS FOR BOTH YOUR TRAP NUMBERS.
YOU GET 6 GUESSES TO GET MY NUMBER.

GUESS # 1 ? 40,70
MY NUMBER IS LARGER THAN YOUR TRAP NUMBERS.

GUESS # 2 ? 80,90
MY NUMBER IS SMALLER THAN YOUR TRAP NUMBERS.

GUESS # 3 ? 73,77
YOU HAVE TRAPPED MY NUMBER.

GUESS # 4 ? 74,76
YOU HAVE TRAPPED MY NUMBER.

GUESS # 5 ? 75,75
MY NUMBER IS SMALLER THAN YOUR TRAP NUMBERS.

GUESS # 6 ? 74,74
YOU GOT IT!!!
```

Look to the Stars

Let's go back to one guess per turn but instead of a clue of "TOO HIGH" or "TOO LOW" now we'll use a clue that tells you roughly how far away you are from the mystery number. The program will not tell you exactly how far away you are -- that would be too easy. Rather, it gives you clues that allow you to determine the range in which the mystery number lies.

EXERCISE - 5

With the same teams, play STARS on the computer. This is more difficult than the previous games; however, with some practice you should be able to come up with a good playing strategy. Efficient playing strategies are very different in STARS than in the other games because of the fairly high information content in the clues. It might help to make up a chart that shows how far you are from the mystery number for a given number of stars. Find a good playing strategy; be prepared to support your method. Play 10 games using your strategy; how many moves on average does it take to win?

```
STARS - A NUMBER GUESSING GAME

DO YOU WANT INSTRUCTIONS (1=YES 0=NO)? 1
I AM THINKING OF A WHOLE NUMBE  FROM 1 TO 100
TRY TO GUESS MY NUMBER. AFTER YOU GUESS, I
WILL TYPE ONE OR MORE STARS (*). THE MORE
STARS I TYPE, THE CLOSER YOU ARE TO MY NUMBER.
ONE STAR (*) MEANS FAR AWAY. SEVEN STARS (*******)
MEANS REALLY CLOSE! YOU GET 7 GUESSES.


OK, I AM THINKING OF A NUMBER. START GUESSING.

YOUR GUESS? 16
*

YOUR GUESS? 88
* * * * *

YOUR GUESS? 92
* * * *

YOUR GUESS? 84
* * * * * *

YOUR GUESS? 82
* * * * * * *

YOUR GUESS? 81
*****************************************************!!!
YOU GOT IT IN  6 GUESSES!!  LET'S PLAY AGAIN...
```

OPTIONAL PROJECT - 1

Make up a number or letter guessing game of your own and write a computer program to play the game.

OPTIONAL PROJECT - 2

Write a computer program to search for a secret number in a list with you giving it clues. In other words, write a program to become the player in GUESS.

OPTIONAL PROJECT - 3

Write a computer program to search for a number (or name if your computer has strings) in a list without a uniform structure. Use the list of telephone exchanges given earlier in this chapter or use the following list.

1	12	33	65
2	14	36	70
3	16	40	75
4	18	44	80
5	21	48	85
6	24	52	90
8	27	56	96
10	30	60	100

The output of the program should look something like this:

```
WHAT NUMBER ARE YOU LOOKING FOR?  60

60 IS THE 24TH NUMBER IN THE LIST OF 32 NUMBERS.
```

Be sure to use a good search technique. Remember, you are writing a program which should work efficiently with a list of thousands of numbers or names. (To write this program you'll need to use subscripted variables for this list. Don't try to do it without; it's much too complicated).

More About Mathematical Logic

Introduction

Most games of chance or games of moderate complexity like checkers do not have known strategies that can guarantee a win or a draw. However, other games give the player total information and have certain optimal playing strategies. From this type of game we can learn a great deal about logical approaches to mathematical problems.

23 Matches

A simple game that is easy to learn and easy to play is 23 Matches. The optimal strategy is based on a simple mathematical relationship, and a computer program that plays a perfect game can therefore easily be constructed.

There are two players and one pile of 23 matches. Each player, in turn, removes any number of matches between 1 and 3. The player who removes the last match loses.

EXERCISE - 1

Divide the class into groups of two. Have the two players in each group play 23 Matches with each other. (You need not use matches; any small object such as buttons, coins, etc. will do as well). Does going first have any advantage? Can you determine a strategy which always guarantees a win?

EXERCISE - 2

Each group of two players who played 23 Matches with each other should play 23MTCH on the computer. The computer always lets you go first. You should always be able to win. Can you?

The easiest way to understand the winning procedure is to start at the end of the game. We must remove from 1 to 3 matches per turn. In order to leave our opponent with 1 match at the end we must make sure that he leaves us with 2, 3 or 4 matches on the next-to-last move. To do this, we must leave him with 5 matches; there is nothing else that will guarantee us getting 2, 3 or 4. Going one step further back, we see that to leave our opponent with 5, we must guarantee that he leaves us 6, 7 or 8. Following the previous logic, we see that we must leave him with 9. And so on.

Notice the relationships. We must always leave our opponent with 1, 5, 9 ... matches, that is, numbers that are 4 apart. This is one greater than the number of matches we can take on each turn. Let's see if we can generalize this role in the next section.

```
RUN 23MTCH
LET'S PLAY 23 MATCHES.  WE START WITH 23 MATCHES.
YOU MOVE FIRST.  YOU MAY TAKE 1,2 OR 3 MATCHES.
THEN I MOVE...I MAY TAKE 1,2 OR 3 MATCHES.
YOU MOVE, I MOVE AND SO ON.  THE ONE WHO HAS TO
TAKE THE LAST MATCH LOSES.
GOOD LUCK AND MAY THE BEST COMPUTER (HA HA) WIN.


THERE ARE NOW 23 MATCHES.

HOW MANY DO YOU TAKE? 1

I TOOK 1 ...THERE ARE NOW 21 MATCHES.

HOW MANY DO YOU TAKE? 3

I TOOK 1 ...THERE ARE NOW 17 MATCHES.

HOW MANY DO YOU TAKE? 1

I TOOK 3 ...THERE ARE NOW 13 MATCHES.

HOW MANY DO YOU TAKE? 2

I TOOK 2 ...THERE ARE NOW 9 MATCHES.

HOW MANY DO YOU TAKE? 4
YOU CHEATED! BUT I'LL GIVE YOU ANOTHER CHANCE.

HOW MANY DO YOU TAKE? 1

I TOOK 3 ...THERE ARE NOW 5 MATCHES.

HOW MANY DO YOU TAKE? 2

I TOOK 2 ...THERE ARE NOW 1 MATCHES.

HOW MANY DO YOU TAKE? 1

I WON!!! BETTER LUCK NEXT TIME.
```

Battle of Numbers

A somewhat more difficult game, but one that uses the same strategy as 23 Matches, is Battle of Numbers. In this game there are two players and one pile of objects, Like 23 Matches, each player, in turn, removes any number of objects from the pile between 1 and N. N is some random number greater than 1 and less than the number of objects in the pile. The player who removes the last object loses.

EXERCISE - 3

Select pairs of two players. Have the players in each pair play Battle of Numbers with each other. Try playing the game with each of the following sets of rules:

Objects in Pile	Remove Each Turn
23	1 to 4
30	1 to 8
17	1 to 2

EXERCISE - 4

Each player pair should play BATNUM on the computer. In this game the computer selects the objects in the pile and the number that can be removed on each turn. Can you determine the formula for the winning numbers to leave your opponent (like 1, 5, 9 ... in 23 Matches)? Your formula should have two variables: M = number of moves from the end of the game, N = maximum number that can be taken on each move.

NIM

Similar to BATNUM in that objects are removed from a pile, NIM differs by having more than one pile, usually three or more. Unlike BATNUM, the player who takes the last piece wins.

EXERCISE - 5

Divide the class into small groups of two or three and have each group play NIM on the computer. Who wins more consistently, computer or humans? Can you determine a winning strategy for playing NIM?

```
THIS PROGRAM PLAYS NIM.
DO YOU WANT INSTRUCTIONS? YES

    NIM IS PLAYED BY TWO PEOPLE PLAYING ALTERNATELY.  BEFORE
THE PLAY STARTS, AN ARBITRARY NUMBER OF STICKS OR OBJECTS IS
PUT INTO AN ARBITRARY NUMBER OF PILES, IN ANY DISTRIBUTION
WHATEVER.  THEN EACH PLAYER IN HIS TURN REMOVES AS MANY
STICKS AS HE WISHES FROM ANY PILE--BUT FROM ONLY ONE PILE,
AND AT LEAST ONE STICK.  THE PLAYER WHO TAKES THE LAST STICK
IS THE WINNER.
    THIS PROGRAM ALLOWS YOU TO SET UP THE INITIAL ARRANGEMENT
OF PILES AND STICKS.  IT WILL NOT ACCEPT MORE THAN TWENTY
PILES OR STICKS IN EACH PILE.

HOW MANY PILES? 4

HOW MANY STICKS IN PILE 1 ? 7
HOW MANY STICKS IN PILE 2 ? 5
HOW MANY STICKS IN PILE 3 ? 3
HOW MANY STICKS IN PILE 4 ? 1

DO YOU WANT TO GO FIRST? YES

WHICH PILE DO YOU WANT STICKS FROM? 1

HOW MANY STICKS? 5

I'LL TAKE 5 STICKS FROM PILE 2 .

PILE NUMBER   STICKS LEFT
 1              2
 3              3
 4              1

WHICH PILE DO YOU WANT STICKS FROM? 3

HOW MANY STICKS? 3

I'LL TAKE 1 STICK FROM PILE 1 .

PILE NUMBER   STICKS LEFT
 1              1
 4              1

WHICH PILE DO YOU WANT STICKS FROM? 1

HOW MANY STICKS? 1

I'LL TAKE 1 STICK FROM PILE 4 .

I WON.  DO YOU WANT TO PLAY AGAIN? NO
```

The discussion of the method of playing NIM is taken from <u>BASIC Programming, Second Edition</u> by Kemeny and Kurtz.

A player who does not know the strategy will find it difficult to win against one who does, and the number of possible combinations is much too large to remember. However, as with BATNUM, there is a mathematical formulation that permits an easy determination of the best move. We shall describe the method and state, without proof, that it does work; a theoretical justification of the method is beyond our goal.

The strategy requires that the number of objects in each pile be represented as a binary number, that is, a number to the base 2. Thus, a configuration of (5, 4, 3) for the number of objects in three piles would be

Decimal	Binary
5	1 0 1
4	1 0 0
3	0 1 1

We shall be concerned with the *parity* of the columns, that is, the evenness or oddness of the number of 1's in each column or bit position. Notice that the first column contains two 1's, and is therefore even. The middle column is odd, since it contains one 1. The third column is again even. The optimal strategy calls for the player (or computer) to remove a number of objects from one pile in order to leave *each* column in an even state. For example, from (5, 4, 3), we would remove 2 from the third pile, leaving (5, 4, 1). The player who takes the last piece (and wins) leaves all the columns with even parity. And it is true that (a) with one or more columns having odd parity, it is always possible to find a move that leaves all columns even, and (b) when all columns are even, any move whatsoever will leave one or more of them odd. Thus, a player who knows the strategy can be sure that, once he produces an all-even position, he can seesaw his way to victory.

It is not always obvious how to make the optimal move. For instance, if the piles contain (26, 9, 7), the optimal move (remove 12 from the first pile) is not readily seen from an inspection of the binary representation

1 1 0 1 0
0 1 0 0 1
0 0 1 1 1.

Furthermore, several different moves may be optimal as, for example, with (7, 6, 5). Removing 4 from *any* pile will leave

(3, 6, 5) or (7, 2, 5) or (7, 6, 1)

all of which are all-even.

The method for determining an optimal move first selects a row having a 1 in the first odd column. Objects are removed from that row so as to change each digit that is also in an odd column from 0 to 1 or 1 to 0, as the case may be.

<u>Even Wins</u>

A game in the same family is called Even Wins. Again there are two players and an odd number pile of objects. Each player, in turn, removes any number of objects from the pile between 1 and N. N is some random number greater than 1 and less than one-quarter of the number of objects in the pile. The player who takes an even number of objects in total wins.

<u>EXERCISE - 6</u>

Divide the class into small groups and play EVEN on the computer. Play a tournament of best 6 out of 11 games with the computer. Who won? Can you determine a consistent winning strategy?

<u>OPTIONAL PROJECT - 1</u>

Write a computer program to play a game of ODD WINS. This is the same game as EVEN WINS except that the player who takes an odd total number of objects wins.

```
EVEN     01:24 PM         13-NOV-73
GAME OF EVEN WINS -- CYBERNETIC VERSION

DO YOU WANT INSTRUCTIONS (Y OR N)? Y

THE GAME IS PLAYED AS FOLLOWS:
AT THE BEGINNING OF A GAME, A RANDOM NUMBER OF CHIPS ARE
PLACED ON THE BOARD.  THE NUMBER OF CHIPS ALWAYS STARTS
AS AN ODD NUMBER.  ON EACH TURN, A PLAYER MUST TAKE ONE,
TWO, THREE, OR FOUR CHIPS.  THE WINNER IS THE PLAYER WHO
FINISHES WITH A TOTAL NUMBER OF CHIPS THAT IS EVEN.
THE COMPUTER STARTS OUT KNOWING ONLY THE RULES OF THE
GAME.  IT GRADUALLY LEARNS TO PLAY WELL.  IT SHOULD BE
DIFFICULT TO BEAT THE COMPUTER TWENTY GAMES IN A ROW.
TRY IT !!!

TO QUIT AT ANY TIME, TYPE '0' AS YOUR MOVE.

THERE ARE 13 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 9     ...YOUR MOVE? 4
THERE ARE 5 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 1     ...YOUR MOVE? 1
GAME OVER ... I WIN !!

THAT WAS FUN!  LET'S PLAY AGAIN...

THERE ARE 19 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 15     ...YOUR MOVE? 2
THERE ARE 13 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 9     ...YOUR MOVE? 2
THERE ARE 7 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 3     ...YOUR MOVE? 2
THERE IS 1 CHIP ON THE BOARD
COMPUTER TAKES 1 CHIP
GAME OVER ... YOU WIN !!

THAT WAS FUN!  LET'S PLAY AGAIN...

THERE ARE 17 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 13     ...YOUR MOVE? 3
THERE ARE 10 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 6     ...YOUR MOVE? 1
THERE ARE 5 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 1     ...YOUR MOVE? 1
GAME OVER ... I WIN !!

THAT WAS FUN!  LET'S PLAY AGAIN...

THERE ARE 19 CHIPS ON THE BOARD.
COMPUTER TAKES 1 CHIP LEAVING 18    ...YOUR MOVE? 4
THERE ARE 14 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 10     ...YOUR MOVE? 4
THERE ARE 6 CHIPS ON THE BOARD.
COMPUTER TAKES 4 CHIPS LEAVING 2     ...YOUR MOVE? 2
GAME OVER ... YOU WIN !!
```

Logic and Inference

Getting Started

How do people solve problems? There are direct and indirect approaches. Frequently, the "scientific method" is advocated as the generalized approach to solving all kinds of problems. Stated briefly, the scientific method consists of the following steps:

1. State the problem. Break it down into manageable pieces, if necessary.
2. Collect all relevant facts and data.
3. Using the data, try a solution. Does it meet the objectives, i.e., does it solve the problem? If so, is it the best solution? If not, go back to Step 2.

But how do we get the solution from the data? Several possibilities exist:

1. Deduction. Reaching a conclusion from something already known.
2. Inference. Reaching a conclusion from facts and evidence.
3. Trial and Error. Keeping at it, avoiding past mistakes until you get it right.
4. Experimentation. Trying something new and observing the results to achieve a goal.
5. Intuition. Direct perception of the truth that bypasses analysis.

In this chapter, we're concerned with deduction and inference. That is not to say that we can't use trial and error, experimentation, and even intuition, but we'll try to focus on the first two.

For example, consider the following problem:

> Four men, one of whom is known to have committed a certain crime, said the following when questioned by an inspector from Scotland Yard.
>
> | | |
> |---|---|
> | Growley: | "Snavely did it." |
> | Snavely: | "Gaston did it." |
> | Gus: | "I didn't do it." |
> | Gaston: | "Snavely lied when he said I did it." |
>
> If only one of the four statements is true, which was the guilty man?

EXERCISE - 1

Solve the problem by whatever means you want. Describe the approach you took. Can your approach be applied to other problems of this type?

BAGELS

In the game of BAGELS the object is to guess a mystery 3-digit number. All three digits are different. After each guess, you are given clues as follows:

PICO - One digit correct but in the wrong place.
FERMI - One digit correct and in the right place.
BAGELS - No digits correct.

Let's say the mystery number is 685. Let's look at a possible sequence of guesses to get the number.

Guess #	Guess	Clue	Discussion
1	123	BAGELS	No digits correct
2	456	PICO PICO	Two digits correct but in in the wrong place. We could assume the 4 and 5 are correct but interchanged and try a new digit for the 6.
3	547	PICO	Oh, oh. We lost a correct digit, but we now know that either the 4 or 5 must go in the last position and we have to bring back the 6.
4	684	FERMI FERMI	Wow! We're getting close. Let's assume the 6 and 4 are both in the correct position, but the 8 is incorrect.
5	694	FERMI	Oh, oh. Since we already know the 6 must be correct from Guess #3, it looks like we've been wrong about the 4 all along. That means (from Guess #4) the 6 and 8 are correct and the other digit must be 5 (from Guess #2).
6	685	YOU GOT IT!	We got it in 6 guesses.

EXERCISE - 2

Play BAGELS in class. Have a student think of a number and write the clues on the blackboard as other students try to guess it.

EXERCISE - 3

Divide the class into teams of four members. Have each team come up with a strategy for playing BAGELS. Have them try their strategy by playing the game on the computer 10 times. What is the average number of guesses for each strategy? Did any groups come up with the same strategy?

```
I AM THINKING OF A THREE-DIGIT NUMBER.  TRY TO GUESS
MY NUMBER AND I WILL GIVE YOU CLUES AS FOLLOWS:
   PICO   - ONE DIGIT CORRECT BUT IN THE WRONG POSITION
   FERMI  - ONE DIGIT CORRECT AND IN THE RIGHT POSITION
   BAGLES - NO DIGITS CORRECT

O.K.  I HAVE A NUMBER IN MIND.
GUESS # 1 ? 123
PICO FERMI
GUESS # 2 ? 421
YOU GOT IT!!!

PLAY AGAIN (1 FOR YES, 0 FOR NO)? 1

O.K.  I HAVE A NUMBER IN MIND.
GUESS # 1 ? 123
PICO
GUESS # 2 ? 415
BAGLES
GUESS # 3 ? 267
PICO
GUESS # 4 ? 892
PICO
GUESS # 5 ? 638
FERMI FERMI
GUESS # 6 ? 639
FERMI
GUESS # 7 ? 738
YOU GOT IT!!!

PLAY AGAIN (1 FOR YES, 0 FOR NO)? 1

O.K.  I HAVE A NUMBER IN MIND.
GUESS # 1 ? 123
PICO
GUESS # 2 ? 415
FERMI
GUESS # 3 ? 617
PICO
GUESS # 4 ? 436
YOU GOT IT!!!

PLAY AGAIN (1 FOR YES, 0 FOR NO)? 0
```

OPTIONAL PROJECT - 1

Write a computer program to simulate the game of BAGELS. There are two quite different algorithms to construct this game. The game in this book uses only one of them, of course. Try to determine the other method.

OPTIONAL PROJECT - 2

Write a computer program to act as the player in the BAGELS game. Use one or more strategies developed in Exercise 3. Using two terminals, play 10 games with this program. What are the average number of guesses? If your computer has a timer function, keep track of the amount of time to do the calculations for 10 games. Compare this with other methods.

OPTIONAL PROJECT - 3

Write a computer program to play BAGELS with more than one of the same digit allowed. How does your guessing strategy change playing this version of BAGELS? Can the same playing strategies be modified for this version or do you need an entirely new approach?

Consider the various tradeoffs in solving the BAGELS problem. There is one player strategy which is extremely trivial consisting of simply guessing every number between 001 and 999. It is simple to program (2 lines) and guarantees a solution in an average of 496 guesses. Another approach (logical inference) guarantees a solution in a maximum of 7 guesses or an average of 4.6 guesses but is quite complex to program. A somewhat simpler approach guarantees a solution in 10 guesses. Comparing, we have:

	. . . Guesses	. . .	
Approach	Maximum	Average	Time to Program
Brute force	987	496	Very short
Simple inference	10	7	Medium
Complex inference	7	4.6	Long

We can see from the table that the inference methods are much more efficient than the linear (or brute force) method. However, if we only intend to solve the problem once or twice, maybe it is not worth while to spend the additional time programming one of the inference methods.

Bulls and Cows

Bulls and Cows (BULCOW) is a 5-digit version of BAGELS. The object of the game is to guess a mystery 5-digit number. All 5 digits are different. After each guess, you are given clues as follows:

1 COW for each digit correct but in the wrong place.
1 BULL for each digit correct and in the right place.

For example,

if the mystery number is	5	0	3	2	7
and your guess is	2	0	1	7	4

your score would be 1 BULL for the 0 and 2 COWS for the 2 and 7.

EXERCISE - 4

Play BULLS and COWS in class. Have a student think of a number and write the clues on the blackboard as other students try to guess it.

EXERCISE - 5

Divide the class into an even number of teams. Pair the teams, say Team A vs Team B, C vs D, etc. Have the teams play BULLS and COWS with each other simultaneously. In other words, Team A and B both think of a number. Team A makes a guess, B gives the score. Then Team B guesses and A gives the score. This goes back and forth until one team wins. The other team must continue until they get the number, however. Teams should play 5 back-and-forth games. Sum the total scores to give an overall team ranking.

EXERCISE - 6

The computer version of BULCOW is designed to play a competitive game as in Exercise 5. Have each student or team play a best 3 out of 5 match with the computer. Who won more matches, computer or students? Why? (The computer uses a good playing strategy, but not the optimum one. Consequently, students should be able to beat it.)

```
     BRADFORD UNIVERSITY BULLS AND COWS GAME

WANT INSTRUCTIONS (Y OR N)? Y

EACH PLAYER (YOU AND THE COMPUTER) THINK OF A 5-DIGIT NUMBER WITH
NO REPEATED DIGITS, E.G., 05387, 24961, 10356.  PLAYERS IN TURN
TRY TO GUESS THE NUMBER OF THE OTHER.  AFTER EACH GUESS, EACH
PLAYER GIVES THE OTHER PLAYER A SCORE FOR HIS GUESS CONSISTING
OF A NUMBER OF BULLS AND COWS.  A BULL IS SCORED FOR EACH CORRECT
DIGIT IN THE CORRECT PLACE.  A COW IS SCORED FOR EACH CORRECT DIGIT
BUT IN THE WRONG PLACE.
     E.G., IF MY NUMBER IS 50327
         AND YOUR GUESS IS 20174
         YOU SCORE 1 BULL (FOR THE 0) AND 2 COWS (FOR THE 2 AND 7)
THE COMPUTER EXPECTS ITS SCORES IN THE FORM '1,2' WHICH WOULD
INDICATE 1 BULL AND 2 COWS.

GOOD LUCK !!



NEW GAME....



YOUR GUESS? 12345
 1 BULL 1 COW - MY GUESS IS 12405 MY SCORE? 0,4
YOUR GUESS? 16278
 1 BULL 1 COW - MY GUESS IS 45721 MY SCORE? 1,3
YOUR GUESS? 46389
 0 BULLS 2 COWS - MY GUESS IS 48512 MY SCORE? 0,4
YOUR GUESS? 60475
 0 BULLS 3 COWS - MY GUESS IS 64251 MY SCORE? 0,4
YOUR GUESS? 18650
 1 BULL 3 COWS - MY GUESS IS 25143 MY SCORE? 3,1
YOUR GUESS? 15960
 1 BULL 4 COWS - MY GUESS IS 25194 MY SCORE? 5,0
 - I WIN - MY NUMBER WAS 19506
```

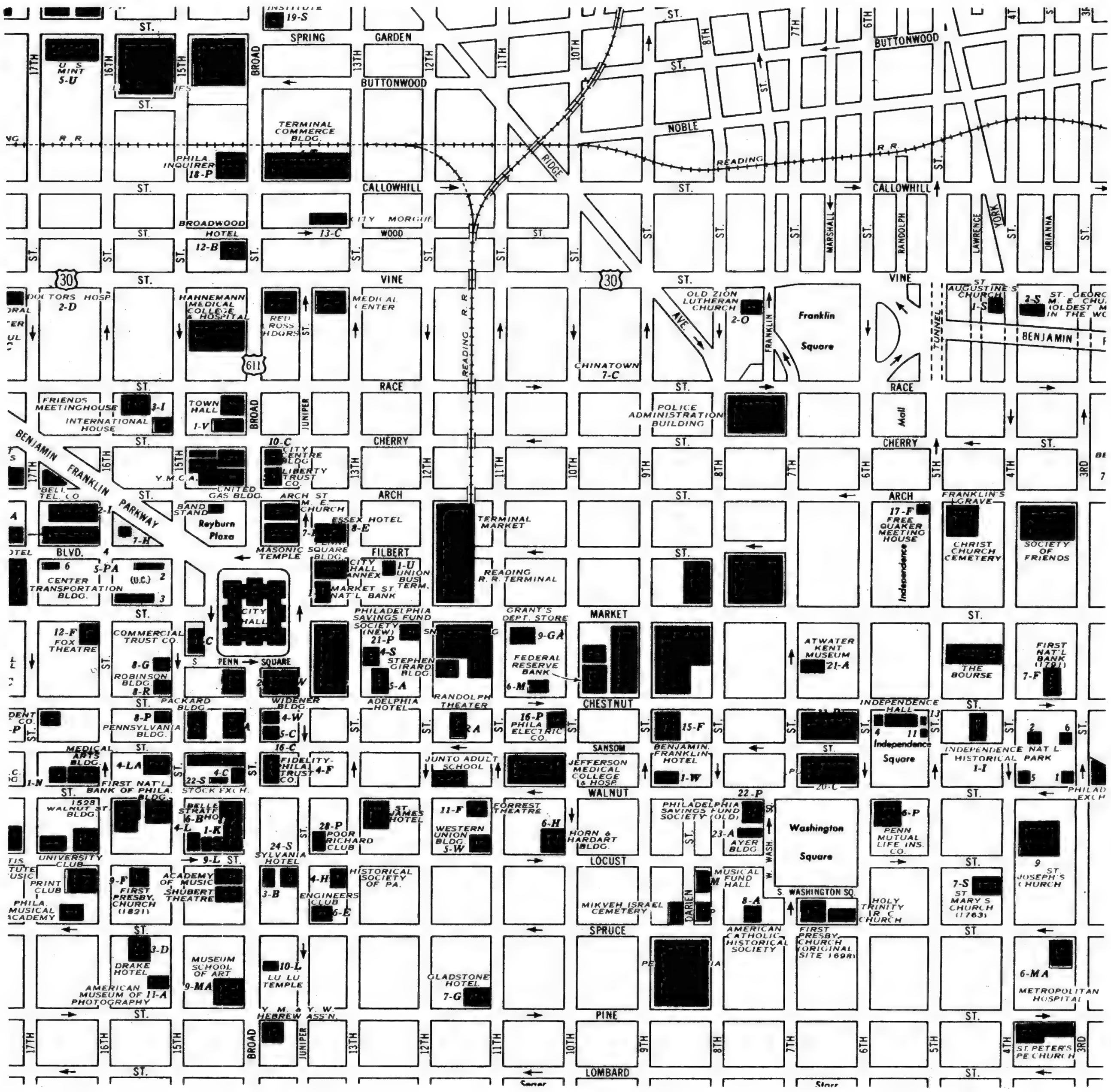
OPTIONAL PROJECT - 4

Write a computer program (or modify BULCOW) to play with more than one digit the same. You may want to cut it down to 4 digits since this is a very difficult game with 5 digits. How does the playing strategy differ from the game with all digits different? How many guesses does it take on average to get the mystery number?

Coordinates and Grids-Getting Started

Introduction

The concept of two-dimensional coordinates are extremely important in many aspects of our lives. For example, say you're at Gimbels (Market and 9th) and want to get to the City Morgue (Wood and 13th). What's the best route? How far must you walk, or should you take a taxi? With the aid of a map we see that Wood is 6 blocks from Market and 13th is 4 blocks from 9th, hence the total distance is 10 blocks. But what if we have a helicopter -- then the distance is only 7.2 blocks. Where did the 7.2 come from? Read on.



Delivering Pizzas

Let's consider an even simpler city than Philadelphia, namely Hyattsville. Only 16 families live in the town, A through P. In the pizza delivery game, you take orders for pizzas. Then you tell a delivery boy where to deliver the ordered pizza. In giving the "address," you always state the horizontal coordinate (across the bottom from left to right) first. Next, you state the vertical coordinate (up the side). This is conventional practice that the horizontal or "X" coordinate is given first followed by the vertical or "Y" coordinate.

Map of Hyattsville

4	M	N	O	P
3	I	J	K	L
2	E	F	G	H
1	A	B	C	D
	1	2	3	4

For example, if G called for a pizza, you would have to tell the driver G's address, which is 3, 2.

EXERCISE - 1

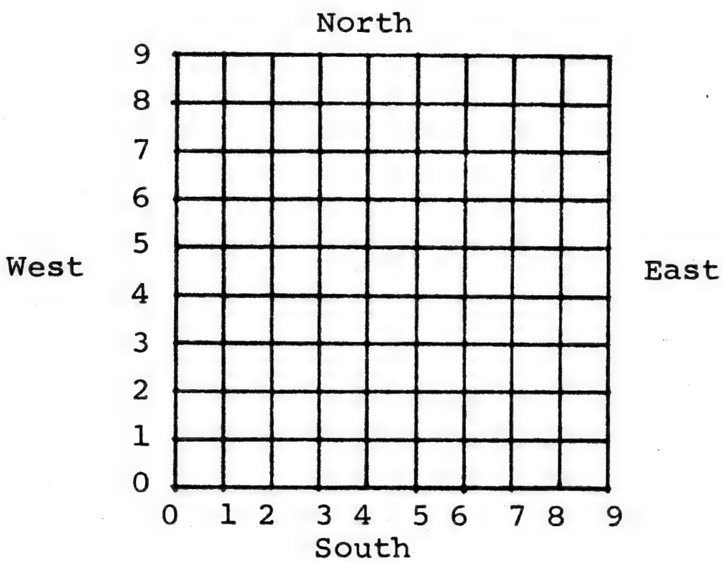
Play the PIZZA game on the computer until you can deliver 5 pizzas in a row without making a mistake.

```
HELLO MICHELOTTI'S PIZZA.   THIS IS L.   PLEASE SEND A PIZZA.
   DRIVER TO MICHELOTTI.   WHERE DOES L LIVE? 3,4
THIS IS O.   I DID NOT ORDER A PIZZA.
I LIVE AT  3 , 4
   DRIVER TO MICHELOTTI.   WHERE DOES L LIVE? 4,3
HELLO MICHELOTTI.   THIS IS L, THANKS FOR THE PIZZA.

HELLO MICHELOTTI'S PIZZA.   THIS IS E.   PLEASE SEND A PIZZA.
   DRIVER TO MICHELOTTI.   WHERE DOES E LIVE? 1,2
HELLO MICHELOTTI.   THIS IS E, THANKS FOR THE PIZZA.

HELLO MICHELOTTI'S PIZZA.   THIS IS A.   PLEASE SEND A PIZZA.
   DRIVER TO MICHELOTTI.   WHERE DOES A LIVE? 1,1
HELLO MICHELOTTI.   THIS IS A, THANKS FOR THE PIZZA.

HELLO MICHELOTTI'S PIZZA.   THIS IS P.   PLEASE SEND A PIZZA.
   DRIVER TO MICHELOTTI.   WHERE DOES P LIVE? 4,4
HELLO MICHELOTTI.   THIS IS P, THANKS FOR THE PIZZA.

HELLO MICHELOTTI'S PIZZA.   THIS IS G.   PLEASE SEND A PIZZA.
   DRIVER TO MICHELOTTI.   WHERE DOES G LIVE? 3,2
HELLO MICHELOTTI.   THIS IS G, THANKS FOR THE PIZZA.

DO YOU WANT TO DELIVER MORE PIZZAS? NO

O.K. MICHELOTTI, SEE YOU LATER!
```

Hunting Hurkles

Now we jump to another galaxy where we're going to hunt Hurkles. Hurkles? A Hurkle is a happy beast that lives on the planet Lirht that has three moons. Hurkles are favorite pets of the Gwik, the dominant race of Lirht and if you really want to know more, get the book A Way Home by Theodore Sturgeon.

Happy Hurkles radiate. Scared Hurkles go invisible. Most of the time they're scared but they want to be found so they'll give you clues where they're hiding. They live in a somewhat larger town than Hyattsville, but still not too big (10 x 10).



You try to guess where the Hurkle is hiding. Remember, horizontal coordinate first, then vertical. After each guess, you get clues of direction. For example:

Guess	Clue
5, 5	Go Northwest
2, 7	Go Northeast
3, 9	Go South
3, 8	You found him!!

EXERCISE - 2

Play Hurkle in class. Have a student decide where the Hurkle is hiding and have other class members guess the location. The student who hid the Hurkle gives clues to the class.

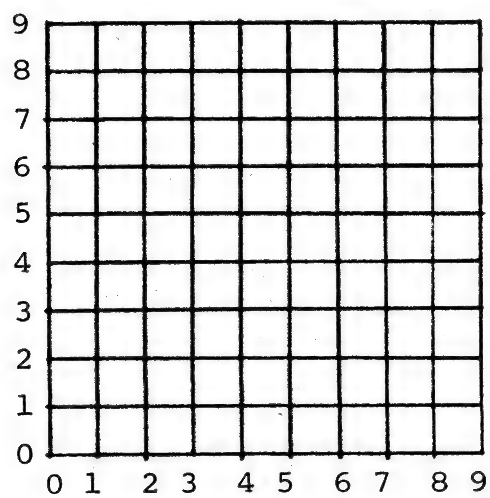
EXERCISE - 3

Divide the class into teams and have them play Hurkle on the computer. Teams should attempt to come up with an optimal guessing strategy. A good strategy should always locate the Hurkle in 5 or fewer guesses. The optimal strategy guarantees finding him in no more than 4 guesses.

Finding Mugwumps

Mugwumps, like Hurkles live in a 10 x 10 town or grid. Unlike Hurkles, four Mugwumps hide out at one time. Also, rather than clues of direction, to locate Mugwumps you are given clues of straight-line distance. (Remember the helicopter in Philadelphia). To measure straight line distance of course we should have a ruler with dimensions the same as those on our grid. As you start to play the game, you might find it's easier to use a compass to measure the distance from the ruler and then draw it on the grid.

On the next page is a sample run of MUGWUMP.



```
RUNNH
THE OBJECT OF THIS GAME IS TO FIND FOUR MUGWUMPS
HIDDEN ON A 10 BY 10 GRID. HOMEBASE IS POSITION 0,0
ANY GUESS YOU MAKE MUST BE TWO NUMBERS WITH EACH
NUMBER BETWEEN 0 AND 9, INCLUSIVE. FIRST NUMBER
IS DISTANCE TO RIGHT OF HOMEBASE AND SECOND NUMBER
IS DISTANCE ABOVE HOMEBASE.

YOU GET 10 TRIES. AFTER EACH TRY, I WILL TELL
YOU HOW FAR YOU ARE FROM EACH MUGWUMP.



TURN NO. 1 WHAT IS YOUR GUESS? 5,5
YOU ARE 2.8 UNITS FROM MUGWUMP 1
YOU ARE 6.4 UNITS FROM MUGWUMP 2
YOU ARE 2.2 UNITS FROM MUGWUMP 3
YOU ARE 2.2 UNITS FROM MUGWUMP 4


TURN NO. 2 WHAT IS YOUR GUESS? 3,3
YOU ARE 4 UNITS FROM MUGWUMP 1
YOU ARE 3.6 UNITS FROM MUGWUMP 2
YOU ARE 1 UNITS FROM MUGWUMP 3
YOU ARE 4.1 UNITS FROM MUGWUMP 4


TURN NO. 3 WHAT IS YOUR GUESS? 4,3
YOU ARE 3 UNITS FROM MUGWUMP 1
YOU ARE 4.4 UNITS FROM MUGWUMP 2
YOU HAVE FOUND MUGWUMP 3
YOU ARE 4 UNITS FROM MUGWUMP 4


TURN NO. 4 WHAT IS YOUR GUESS? 7,3
YOU HAVE FOUND MUGWUMP 1
YOU ARE 7.2 UNITS FROM MUGWUMP 2
YOU ARE 5 UNITS FROM MUGWUMP 4


TURN NO. 5 WHAT IS YOUR GUESS? 4,7
YOU ARE 7.2 UNITS FROM MUGWUMP 2
YOU HAVE FOUND MUGWUMP 4


TURN NO. 6 WHAT IS YOUR GUESS? 0,1
YOU HAVE FOUND MUGWUMP 2

YOU GOT THEM ALL IN 6 TURNS!

THAT WAS FUN! LET'S PLAY AGAIN.....
FOUR MORE MUGMUMPS ARE NOW IN HIDING.


TURN NO. 1 WHAT IS YOUR GUESS? ^C

READY
```

EXERCISE - 4

Divide the class into teams and have each team play MUGWUMP 3 times on the computer. Are any guessing approaches better than others? Try playing one game without using a grid diagram. Nearly impossible, isn't it?

OPTIONAL PROJECT - 1

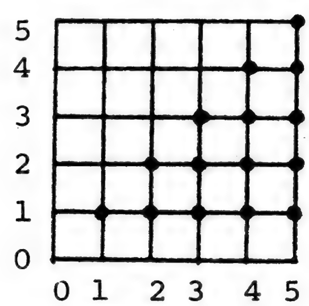
Modify the MUGWUMP program to change the grid size and number of guesses allowed. How many guesses does it take on average to find the Mugwumps on a 5 x 5 grid, 10 x 10 grid, and 20 x 20 grid?

OPTIONAL PROJECT - 2

Diagonal distance between two points on a grid (or matrix) can be computed with the formula:

$$d = \sqrt{x^2 + y^2}$$

in which d is the distance, x is the horizontal distance and y is the vertical distance. Say you're at(0, 0)on a grid. Prepare a table that shows all the diagonal distances to Points (1,1), (2,1), (2,2), (3,1), (3,2), (3,3), (4,1), (4,2), (4,3), (4,4), (5,1), (5,2), (5,3), (5,4), and (5,5). Play a game of MUGWUMP now using this table instead of a ruler.



Coordinates and Grids-Moving Along

Getting Oriented

When following a road map, it is easiest if we align the map with the terrain, i.e., map North to real North. However, it may not always be possible to do this. Sometimes we are lost so we don't know which way is North. Or we may be in a location that doesn't fall on the map. Or the map may not contain enough information to orient it correctly. In all these instances, our first objective should be to obtain the information needed so we can orient the map correctly.

To orient a map, we must make comparisons between the map and terrain until we get a match. This is something like trying to crack a code. For example, there is an alphabet code which codes "MARY" as "NBSZ" and "BOB" as "CPC." What is the code? Obviously you just take the next letter. How did you figure that out? By comparing one to the other, of course. Remember this when we start to decode fleet dispositions.

Decoding an Enemy Fleet Disposition

The computer game BATTLE is based on the popular game Battleship. The primary purpose of the game is to familiarize one with the location and designation of points on a grid and make comparisons using this information.

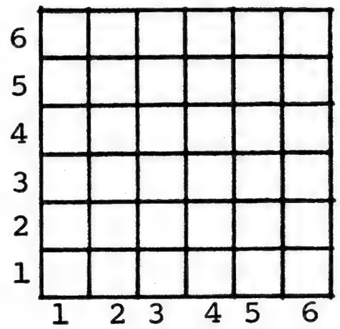
The BATTLE program first sets up the enemy fleet disposition. The fleet consists of 6 ships: two destroyers (ships number 1 and 2) which are two units long, two cruisers (3 and 4) which are three units long, and two aircraft carriers (5 and 6) which are four units long. The program then prints out this fleet disposition in a coded or disguised format as shown below.

```
THE FOLLOWING CODE OF THE ENEMY FLEET DISPOSITION
HAS BEEN CAPTURED BUT NOT DE-CODED:

3     0     0     0     1     1
3     0     0     0     0     0
3     2     6     6     6     6
4     0     2     0     0     0
4     5     5     5     5     0
4     0     0     0     0     0

DE-CODE IT AND USE IT IF YOU CAN BUT
KEEP THE DE-CODING METHOD A SECRET.
```

Ship number 1 is a destroyer.

Ship number 5 is an aircraft carrier.

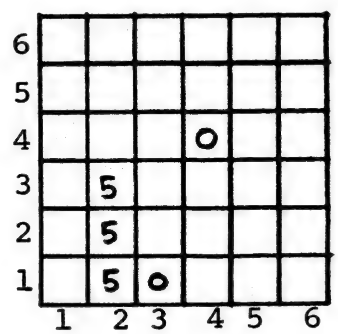
You then try to sink the various ships by typing in the coordinates (two digits, each from 1 to 6, separated by a comma) of the place you want to drop a shell. The computer gives the appropriate response (splash or hit) which you should record on a 6 x 6 grid.



You are thus building a representation of the actual fleet disposition with which you can hopefully decode the original coded fleet disposition printed out by the computer.

Here are the first five moves from an actual game of BATTLE. We used a 0 to designate a splash (or location of no ship).

```
START GAME
? 3,1
SPLASH!  TRY AGAIN.
? 2,1
A DIRECT HIT ON SHIP NUMBER 5
TRY AGAIN.
? 2,2
A DIRECT HIT ON SHIP NUMBER 5
TRY AGAIN.
? 2,3
A DIRECT HIT ON SHIP NUMBER 5
TRY AGAIN.
? 4,4
SPLASH!  TRY AGAIN.
?
```



EXERCISE - 1

Each student (or student teams) should play BATTLE on the computer until he can successfully decode the enemy fleet disposition.

Salvo

The computer game SALVO is a simulation of the board game Battleship. The game is played on a 10 x 10 grid. Two players (you and the computer) each have four ships: a battleship (5 units long), cruiser (3 units long), and two destroyers (2 units long). The ships may be placed horizontally, vertically or diagonally and must not overlap. Your ships and those of the computer are located on two different 10 x 10 grids; think of them as adjacent portions of the ocean since you fire back and forth at each other.

On each turn you fire shots at your opponent. Each ship is able to fire as follows:

Battleship	3 shots
Cruiser	2 shots
Destroyer (2)	1 shot

Hence, you get up to 7 shots per turn. If the enemy sinks your cruiser you have only 3+1+1=5 shots. As in BATTLE and the other previous grid games, you enter a shot by first giving its horizontal (x) coordinate, a comma, and then the vertical (y) coordinate.

Keep a record of your shots fired at the enemy on a 10 x 10 grid. You may, but don't have to, keep a record of enemy shots fired at you.

This game is somewhat more difficult than BATTLE since you enter all 7 (or fewer) of your shots before you learn where they landed. Furthermore, you are not told which shot landed on what, just a summary. Consequently, you must devise a way to keep tentative records of the location of the enemy ships.

EXERCISE - 2

Divide the class into teams. Each team should play SALVO on the computer in a best two out of three battles. Who won most, student teams or the computer? The computer plays a good, but by no means optimum strategy. It should be possible to beat but it's not a pushover.

```
ENTER COORDINATES FOR...
BATTLESHIP
? 1,1
? 1,2
? 1,3
? 1,4
? 1,5
CRUISER
? 9,1
? 9,2
? 9,3
DESTROYER<A>
? 9,9
? 8,8
DESTROYER<B>
? 1,9
? 2,8
DO YOU WANT TO START? Y
```

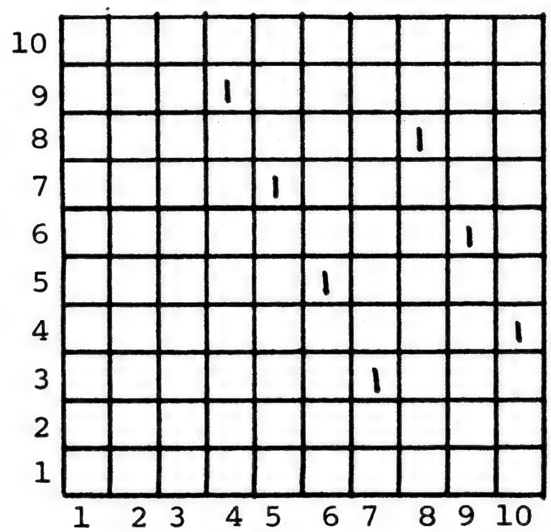
```
TURN 4

YOU HAVE 7 SHOTS
? 7,1
? 5,3
? 9,4
? 9,6
? 3,3
? 5,1
? 7,8
I HAVE 7 SHOTS
 1   6
 1   8
 1   5
 3   5
 1   10
 2   9
 3   10
I HIT YOUR BATTLESHIP
```

```
TURN 5

YOU HAVE 7 SHOTS
? 10,6
? 9,1
? 6,8
? 4,10
? 6,10
? 7,9
? 8,7
YOU HIT MY DESTROYER<A>
I HAVE 7 SHOTS
 4   9
 4   10
 2   4
 4   4
 4   6
 1   4
 4   1
```

EXERCISE - 3

Evaluate various playing strategies. (Incidentally, SALVO can be played between two teams or two players once you get the hang of it. This may be desirable if there is a shortage of terminal time.). One strategy places a premium upon locating the enemy battleship and concentrating salvos to sink it in a hurry.

What are the best opening strategies? One approach is to quarter the board and put the first four salvos in each quarter dispersed as widely as possible in the quarter. Another approach is to put one shot in each of several adjacent lines covering groups of lines in all three directions such as shown below. Which strategy seems to be consistently best?

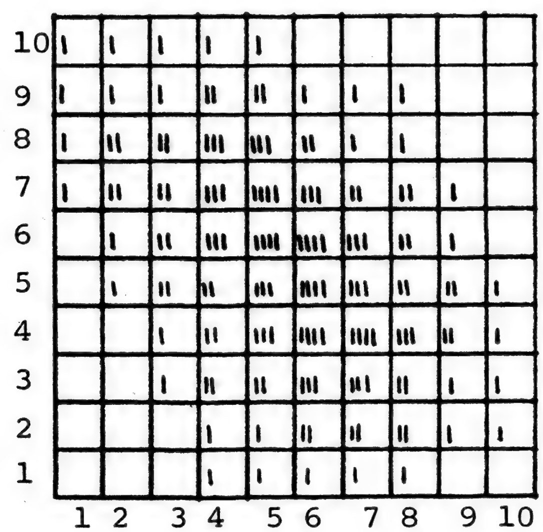


As a converse to the firing strategy, where is the best place to put your ships to escape first round hits? Or to survive the longest?

OPTIONAL PROJECT - 1

Determine a mathematically rigorous value for each square on the grid -- call it a measure of SAFETY to locate your ships. You should use the basic assumption that your opponent is an intelligent player and is going to consistently use the salvo pattern diagrammed in Exercise 3. However, he is just as likely to place it in one orientation as any other. We can sum up the number of times each square is hit if every possible orientation is used once, and this sum is the measure of the probability that a particular square will be hit.

An easy method of operation is as follows. From a piece of paper, cut out the 7 squares in the chosen formation. Using this as a mask or "grille," lay it on a 10 x 10 grid, move it vertically and horizontally to every possible position within the grid, and in each position mark a tick through each of the 7 holes onto the paper below. Rotate the grill 90°, 180°, and 270° and repeat. Turn the grill over and repeat again. The final result shows how many times each call is hit if every possible orientation is used once. The diagram below shows the result of the first 20 positions.



OPTIONAL PROJECT - 2

Write a computer program to act as one player in SALVO, i.e., to determine where to fire but not where to locate ships. Pit this program against SALVO on the computer or in a manual game. What kind of strategy of play works best? Is it best to stick with one strategy or alter your strategy as it gets later into the game?

Abstract Models

Introduction

Generally, a mathematical model is a representation of some real-life process, expressed in mathematical form (such as a set of related equations) or in algorithmic form (such as a computer program). Usually the model is by necessity a simplification of the actual process, since real-life processes tend to be highly complex. One advantage of embodying the model as a computer program is that we can run the program and thus simulate the process being modelled. By varying certain features of the program, we can learn something about the relationships between the components and the overall structure of the process. In addition, if the output does not sufficiently coincide with observed reality, the model can be revised and improved.

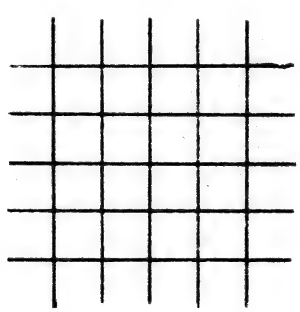
It is also possible to model a purely abstract process. We don't often see this done. After all, if someone asked you to describe some abstract process, what would you say? However, many games start out as purely abstract processes. For example, tic-tac-toe or checkers are abstract from the point of view that they represent no real-life process. Occasionally, it turns out that an abstract process represents a real-life process either by accident or design. The model described below is one such abstract game that in some ways actually represents life itself.

The Game of Life

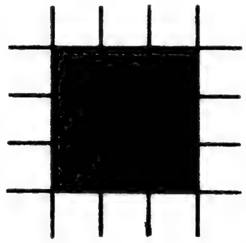
The Game of "Life" was devised by John Conway, a mathematician at the University of Cambridge, and made popular by a series of articles written by Martin Gardner in recent issues of Scientific American. Ever since the first article appeared in October 1970, hundreds of mathematicians throughout the world have become fascinated with the model and have been exploring its properties.

The game consists of following the successive generations of a particular imaginary type of cellular life-form. The life processes of these cells are represented by the following mathematical model, which captures several properties common to all life-forms.

(1) World -- Cells live on an infinite two-dimensional plane of squares (like an infinite checker-board except that all squares are identical).



(2) Neighborhood -- Each square has eight "neighbor" squares. In the diagram below, the neighbor squares for the square with the asterisk (*) have been colored in.



(3) Survival -- A cell (always represented by a *) which is living in generation n, will remain living in generation n+1 if and only if it has exactly 2 or 3 living neighbors in generation n.

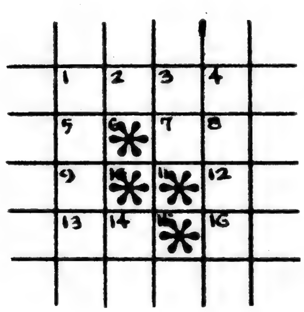
(4) Death -- However, in all other cases the cell dies. Specifically: If it has 0 or 1 neighbors it dies from isolation. If it has 4, 5, 6, 7 or 8 neighbors, it dies from overpopulation.

(5) Birth -- If a square is empty during generation n, a living cell will be born into that square during generation n+1 if and only if that square had exactly 3 (no more, no less) living neighbors during generation n.

The only trick is to remember that all survivals, deaths, and births occur simultaneously, and so the simplest way to keep the "bookkeeping" straight is to have two separate copies of the world -- one for the old generation and one for the new one you are forming. For each square in the old world, decide what its state will be next time, and mark this down in the corresponding square in the new world.

The game is played simply by picking some initial starting pattern and watching the development of some very interesting, and often beautiful, patterns of symmetry. However, the player must be extremely careful because mistakes are easy to make.

As an example, we will trace three generations of the following initial pattern (we have numbered some rows and columns for reference purposes only):



Following the rules of our model:

No births will occur in squares 1, 2, 3, 4 or 5 because none has three living neighbors

The cell in square 6 will survive because it has two living neighbors (10 and 11)

A birth occurs in square 7 because there are three living neighbors (6, 10 and 11)

No birth occurs in squares 8 or 9

The cell in square 10 survives because it has three living neighbors (6, 11 and 15)

The cell in square 11 survives also because it has three living neighbors (6, 10 and 15)

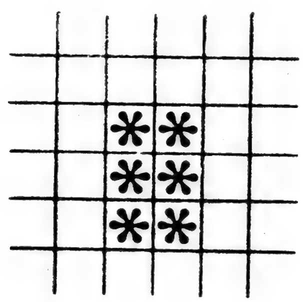
No birth occurs in squares 12 or 13

A birth occurs in square 14 because there are three living neighbors (10, 11 and 15)

The cell in square 15 survives because it has two living neighbors (10 and 11)

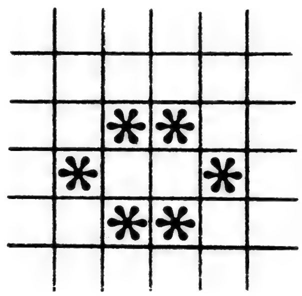
No birth occurs in square 16 because it only has two living neighbors

During this process, we have been filling in a picture of G1, and the end result is:



EXERCISE - 1

Using pencil and paper, carefully compute G2, the next generation for this same society of cells. If you do it correctly, you will find that G2 is the pattern which Conway calls the "beehive":

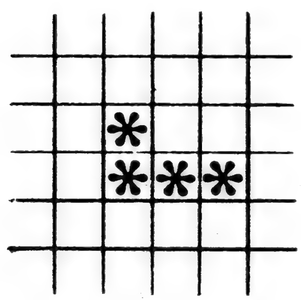


EXERCISE - 2

Now compute G3. If you are again careful, you will discover that G3 is identical to G2. Why does Conway call the beehive a "still-life"? If you are not sure, think about G4, G5, G6. . .

EXERCISE - 3

Using pencil and paper, compute G0, G1, G2 and G3 for the initial pattern below:



If you do it correctly, G3 should look familiar to you.

PROJECT - 1

By now you've no doubt noticed that with pencil and paper, this game is an extremely slow process, and mistakes are all too common. If we ever hope to look at more than a few patterns, we're going to have to turn to the computer for help.

Write a computer program which simulates "Life" for <u>any</u> given initial pattern, and which has the following features:

(1) Allow for as large a world size as your particular computer facility will permit (obviously an infinite plane is not possible in a finite memory). You will probably want to use array structures with two subscripts (row and column).

(2) Whatever world-size you are limited to, make sure your program doesn't try to allow births outside your world, even though properly these <u>would</u> occur on an infinite plane.

(3) Make sure your algorithm allows all survivals, deaths, and births during a given generation to occur <u>simultaneously</u>, as discussed above.

(4) Allow the user to input the initial pattern in a convenient format, such as pairs of (row, column) coordinates.

(5) Make your program efficient and your output as close to the format of the pictures above as possible.

Once your algorithm is designed, and your program is written, debug your program by running it on the following initial G0 pattern, and carefully check your output vs. the results below:

```
G0    G1    G2    G3    G4    G5

             *     *            *
 *    ***   * *    *    ***    * *
***   ***         * *   * *   *   *   etc.
       *    ***    *    ***    * *
                   *            *
```

<u>Warning!</u>

Depending upon the world-size you are limited to, certain "large" patterns may grow differently than they would on an infinite plane.

If the society of cells above, however, fits inside your world-size, you will notice an interesting cyclic pattern beginning at G0, which Conway calls "traffic lights".

PROJECT - 2

When your program is thoroughly debugged and operational, or using the existing LIFE program, the real fun comes in thinking up initial patterns and watching them grow. Interesting situations to watch for are:

(1) Other "still-life" societies (like the "bee-hive")

(2) Other "cyclic" societies (like the "traffic lights")

(3) A society which lives for an extended period of time without dying, becoming still, or cycling

Bring your findings to class for comparison and discussion.

OPTIONAL PROJECT - 3

Find copies of the October 1970 and/or February 1971 issues of Scientific American and read Gardner's articles on "Life." You may want to run your program on some of the societies he describes, such as: diagonal chains, the R pentomino, the Latin cross, the cheshire cat, and many others. Your school library may subscribe to the magazine. If not, the public library near you will certainly have it available.

OPTIONAL PROJECT - 4

Try to think up changes in the model (and your computer program) which will drastically alter the life patterns of the cells, i.e. by modifying the rules for birth or death or both. Based upon your experience so far, try to come up with sets of rules which will lead to more populous societies, or more sparse societies, or societies which are less symmetric than those of "Life", etc. The range of possibilities is very large. Bring your findings to class for comparison and discussion.

OPTIONAL PROJECT - 5

Make some major modifications in your computer program to make it more general, by allowing the user to specify the particular model he wants to investigate. For example, you might have your program begin by posing the following questions to the user:

How many neighbors for survival?

How many neighbors for birth?

Then, if the user answered 2, 3 for the first question and 3 for the second, your program would follow the rules of "Life." But if he gave other answers, the program would simulate for him some other model he wants to investigate.

OPTIONAL PROJECT - 6

Is there any way you can streamline your program so that you can enlarge the size of the world it currently handles?

OPTIONAL PROJECT - 7

How might you alter the general concept of "neighborhood" so that entirely different models could be tested? How would your computer program have to be changed in order to simulate these new models.

OPTIONAL PROJECT - 8

It is possible to place two different types of life forms in the world and let them interact with each other subject to the standard rules for LIFE. A competitive LIFE game, LIFE-2, has been devised which allows two players to put 3 life forms on the board per move and try to eliminate the opposing player. Class members may play this game as individuals or as teams. Are there any playing strategies which can be rigorously stated? Can two good players go on forever with no winner?

Potpourri

Probability

This started out to be an entire chapter on probability. However, that seems to be a favorite subject of just about every other author of a textbook on BASIC. Consequently, we'll dispense with the background discussion and just include a couple of suggested exercises.

EXERCISE - 1

Run the program DICE on the computer. This program "rolls" a pair of dice any number of times you want and shows the number of rolls of each number total. Determine a mathematical prediction for the number of times each number should come up (2 to 12) for a given number of rolls. It might help to make up a table like this:

Score	Combinations that Produce this Score	Number of Combinations	% of Total
2	1-1	1	
3	1-2, 2-1	2	
4	1-3, 2-2, 3-1	3	
etc.			

EXERCISE - 2

In small groups, play CRAPS on the computer. What is the probability of losing on the first roll, what is the probability of "making your point?" Overall, what are the odds in favor of winning (or losing) at CRAPS? Can you determine the overall odds for the house?

EXERCISE - 3

Card games such as Poker are somewhat similar to dice games. In standard draw Poker, your hand is equally likely to contain any of the 52 cards in the deck for your 5 cards. Play POKER on the computer. Play 21 hands and keep track of the cards you got on the original deal of each hand. Theoretically you should have gotten each card twice, however, this isn't nearly enough trials to get a good distribution. Compare your actual hands with the expected. You should have gotten at least 7 pairs in in 21 hands on the first deal -- did you? What else did you get? Compare this with other groups in the class.

OPTIONAL PROJECT - 1

Determine by hand or with a computer program the theoretical probability of getting each poker hand on the first draw. Hands are:

Royal Flush	Two Pair
Flush	Full House
Five of a kind	Straight
Four of a kind	Pair
Three of a kind	Nothing

EXERCISE - 4

Play Blackjack on the computer. A number of years ago, some computer programmers came up with a strategy for playing Blackjack and won thousands of dollars in Las Vegas during a Fall Joint Computer Conference. As a result, the house rules for playing Blackjack were modified slightly. However, it is still possible to consistently win if and only if you stick to an optimum strategy. This strategy is fairly complex and it is beyond the scope of this book to describe it. (A summary was published in Datamation a couple of years ago). We'll just pose one interesting question. The house is required to hit on 16. That means a card with face value of 1 to 5 will keep the house total under 21 while a card with a value of 6 or over will bust the house. Only 5/13 of the cards in the deck have a value of 5 or under while 8/13 have a value over 5. This seems to imply that the house should lose in a ratio of 8 to 5. Does it? Why or why not?

Heuristics

A heuristic approach to solving a problem can be thought of as a rule of thumb. Some rules of thumb are very good and lead to good solutions, others are not as good. Consequently, using a heuristic approach doesn't guarantee the best possible solution but for very complex problems (and even some simple ones) it may be a more efficient approach than a rigorous linear or mathematical method which guarantees a perfect solution.

The science of heuristic problem solving using the computer has become very advanced and is widely used for things like locating warehouses, railroad car routing and other problems involving hundreds of variables and many alternative solutions. Consider: a linear programming solution to routing a mixed load boxcar from Boston to receiving points in Hartford, Columbus, Atlanta, and Baton Rouge would take about 0.72 hours to run on a computer. The heuristic solution takes 0.002 seconds to run, yet it generally yields a solution within 5% of the linear (perfect) solution. Obviously, with millions of cars to be routed every day, the linear approach is not economically feasible.

EXERCISE - 5

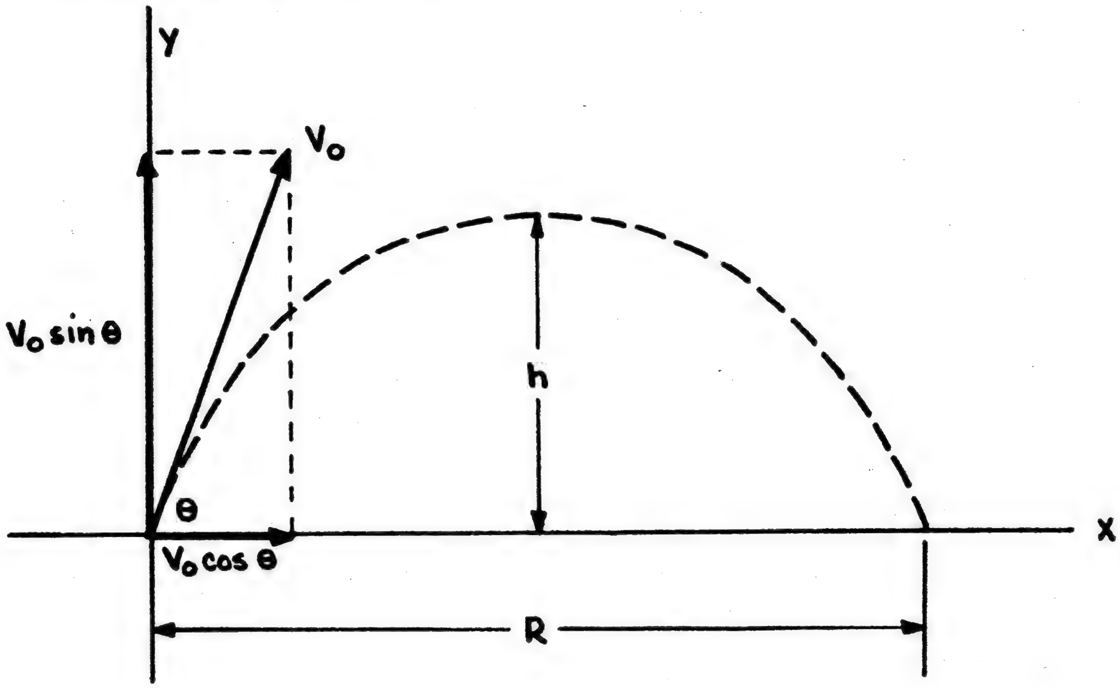
The game of REVERSE lends itself very well to a heuristic approach. There are many possible solutions to each game. One is best, but the mathematics to determine this solution are quite complex and would be extremely time-consuming to calculate. (There is a simpler algorithmic solution but it is not optimal.) A good heuristic approach generally yields a solution within 1 or 2 moves of the perfect solution, i.e., within 10 to 20% of perfection. Divide up into groups of 2 to 4 people. Each group should play REVERSE on the computer at least 10 times. Can you determine a heuristic strategy that solves the puzzle in an average of 10 or fewer moves. You may well use more than one rule of thumb (heuristic). Use the first few games to try things out. Try to describe your heuristics as concisely as possible. Compare your heuristics with those of other groups.

```
REVERSE -- A GAME OF SKILL

WANT THE RULES (1 FOR YES, 0 FOR NO)? 1

THIS IS THE GAME OF REVERSE.  TO WIN, ALL YOU HAVE
TO DO IS ARRANGE A LIST OF NUMBERS (1 THROUGH 9 )
IN NUMERICAL ORDER FROM LEFT TO RIGHT. TO MOVE, YOU
TELL ME HOW MANY NUMBERS (COUNTING FROM THE LEFT) TO
REVERSE. FOR EXAMPLE, IF THE CURRENT LIST IS:

2  3  4  5  1  6  7  8  9

AND YOU REVERSE 4, THE RESULT WILL BE:

5  4  3  2  1  6  7  8  9

NOW, IF YOU REVERSE 5, YOU WIN!

1  2  3  4  5  6  7  8  9

NO DOUBT YOU WILL LIKE THIS GAME OF SKILL, BUT
 IF YOU WANT TO QUIT, REVERSE 0 (ZERO).

HERE WE GO ... THE LIST IS:

 3  7  6  5  4  1  9  2  8

HOW MANY SHALL I REVERSE? 9

 8  2  9  1  4  5  6  7  3

HOW MANY SHALL I REVERSE? 2

 2  8  9  1  4  5  6  7  3

HOW MANY SHALL I REVERSE? 3

 9  8  2  1  4  5  6  7  3

HOW MANY SHALL I REVERSE? 9

 3  7  6  5  4  1  2  8  9

HOW MANY SHALL I REVERSE? 5

 4  5  6  7  3  1  2  8  9

HOW MANY SHALL I REVERSE? 4

 7  6  5  4  3  1  2  8  9

HOW MANY SHALL I REVERSE? 7

 2  1  3  4  5  6  7  8  9

HOW MANY SHALL I REVERSE? 2

 1  2  3  4  5  6  7  8  9

YOU WON IT IN 8 MOVES !!!

TRY AGAIN (1 FOR YES, 0 FOR NO)? 0

O.K.  HOPE YOU HAD FUN!!
```

Projectile Motion

The path followed by a projectile is called its trajectory. The trajectory is affected to a large extent by air resistance, which makes an exact analysis of the motion extremely complex. We shall, however, neglect the effects of air resistance and assume the motion takes place in empty space.

In the general case of projectile motion, the body (bullet, rocket, mortar, etc.) is given an initial velocity at some angle θ above (or below) the horizontal.



If V_o represents the initial velocity (muzzle velocity), the horizontal and vertical components are:

$$V_{ox} = V_o \cos\theta \ , \ V_{oy} = V_o \sin\theta$$

Since we are neglecting air resistance, the horizontal velocity component remains constant throughout the motion. At any time, it is:

$$V_x = V_{ox} = V_o \cos\theta = \text{constant} \tag{1}$$

The vertical part of the motion is one of constant downward acceleration due to gravity. It is the same as for a body projected straight upward with an initial velocity $V_o \sin\theta$. At a time t after the start, the vertical velocity is:

$$V_y = V_{oy} - gt = V_o \sin\theta - gt \tag{2}$$

where g is the acceleration due to gravity.

The horizontal distance is given by:

$$x = V_{ox}t = (V_o \cos\theta)\, t \tag{3}$$

and the vertical distance by:

$$y = V_{oy}t - 1/2\, gt^2 = (V_o \sin\theta)\, t - 1/2\, gt^2 \tag{4}$$

The time for the projectile to return to its initial elevation is found from Equation (4) by setting y = 0. This gives

$$t = \frac{2\, V_o \sin\theta}{g} \tag{5}$$

The horizontal distance when the projectile returns to its initial elevation is called the horizontal range, R. Introducing the time to reach this point in Equation (3), we find:

$$R = \frac{2\, V_o \sin\theta \cos\theta}{g} \tag{6}$$

Since $2\sin\theta \cos\theta = \sin 2\theta$, Equation (6) becomes:

$$R = \frac{V_o^2 \sin 2\theta}{g} \tag{7}$$

The horizontal range is thus proportional to the square of the initial velocity for a given angle of elevation. Since the maximum value of $\sin 2\theta$ is 1, the maximum horizontal range, Rmax is v_o^2/g. But if $\sin 2\theta = 1$, then $2\theta = 90°$ and $\theta = 45°$. Hence the maximum horizontal range, in the absence of air resistance, is attained with angle of elevation of 45°.

From the standpoint of gunnery, what one usually wishes to know is what the angle of elevation should be for a given muzzle velocity v_o in order to hit a target whose position is known. Assuming the target and gun are at the same elevation and the target is at a distance R, Equation (7) may be solved for θ.

$$\theta = 1/2 \sin^{-1}\left(\frac{Rg}{V_o^2}\right) = 1/2 \sin^{-1}\left(\frac{R}{R_{max}}\right) \tag{8}$$

Provided R is less than the maximum range, this equation has two solutions for values of θ between 0° and 90°. Either of the angles gives the same range. Of course, time of flight and maximum height reached are both greater for the high angle trajectory.

For example, say the maximum range of our gun is 10,000 yards and the target is at 5,900 yards.

$$\theta = 1/2 \sin^{-1} \frac{5{,}900}{10{,}000}$$

$$= 1/2\ 36°$$

$$= 18°, \text{ or } 90° - 18° = 72°$$

EXERCISE - 1

Play the computer game GUNNER. Use trial and error to try and destroy the target (see sample run). You get 5 chances per target. How many shots did it take to destroy all 5 targets? Did you ever fail to destroy a target in 5 trials?

EXERCISE - 2

Now play GUNNER but determine the correct firing angle from either sine tables or a slide rule. You should be able to destroy every target with just one shot. What happens when the target is very close? Can you always use whole angles?

EXERCISE - 3

Write a computer program to accept the maximum range of your gun and the range to the target and then calculate the correct firing angle. You will have to solve two problems to write such a program:

1. The BASIC language does not have an ARCSIN function. However, the following formula may help:

$$\sin^{-1} x = \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$$

2. You must convert from radians to degrees.

EXERCISE - 4

Play the game GUNER1 which substitutes a moving target for the stationary one in GUNNER. How does your strategy change? Is it useful to use a method other than trial and error to play the game?

```
THIS COMPUTER DEMONSTRATION SIMULATES THE
RESULTS OF FIRING A FIELD ARTILLERY WEAPON.

YOU ARE THE OFFICER-IN-CHARGE, GIVING ORDERS TO THE GUN
CREW, TELLING THEM THE DEGREES OF ELEVATION YOU ESTIMATE
WILL PLACE THE PROJECTILE ON TARGET.  A HIT WITHIN 100 YARDS
OF THE TARGET WILL DESTROY IT.  TAKE MORE THAN 5 SHOTS,
AND THE ENEMY WILL DESTROY YOU!

MAXIMUM RANGE OF YOUR GUN IS 46500 YARDS.

     DISTANCE TO THE TARGET IS 42995 YARDS.....


ELEVATION:? 40
OVER TARGET BY 2798 YARDS.

ELEVATION:? 37
OVER TARGET BY 1703 YARDS.

ELEVATION:? 34
OVER TARGET BY 118 YARDS.

ELEVATION:? 33.8
***TARGET DESTROYED***    4 ROUNDS OF AMMUNITION EXPENDED

THE FORWARD OBSERVER HAS SIGHTED MORE ENEMY ACTIVITY.
     DISTANCE TO THE TARGET IS 17673 YARDS.....


ELEVATION ? 14
OVER TARGET BY 4156 YARDS.

ELEVATION:? 11
SHORT OF TARGET BY 255 YARDS.

ELEVATION:? 11.2
***TARGET DESTROYED***    3 ROUNDS OF AMMUNITION EXPENDED

THE FORWARD OBSERVER HAS SIGHTED MORE ENEMY ACTIVITY.
     DISTANCE TO THE TARGET IS 41083 YARDS.....


ELEVATION:? 31
***TARGET DESTROYED***    1 ROUNDS OF AMMUNITION EXPENDED

THE FORWARD OBSERVER HAS SIGHTED MORE ENEMY ACTIVITY.
     DISTANCE TO THE TARGET IS 19438 YARDS.....


ELEVATION:? 12.4
***TARGET DESTROYED***    1 ROUNDS OF AMMUNITION EXPENDED

THE FORWARD OBSERVER HAS SIGHTED MORE ENEMY ACTIVITY.
     DISTANCE TO THE TARGET IS 38879 YARDS.....


ELEVATION:? 28.3
***TARGET DESTROYED***    1 ROUNDS OF AMMUNITION EXPENDED
```

<u>Sample run of GUNNER</u>

The first two targets we used trial and error. The last three we used a slide rule. Some difference, eh?

Postlogue

This is not the end, it is only a beginning. As Kenneth Boulding said, "The only thing certain about the future is uncertainty." Hence, the best way to prepare for the future is to be prepared to be surprised. Think about ways of using computers and computer games to produce people who are adaptable.

Experiment with other games in 101 BASIC Computer Games.

Make up a new game. Write a computer program to play it.

Think up a game using colors. Using the contents of your purse or glove compartment. Using Eskimo tools. Using the ingredient list of Rice Krispies.

Experiment! Create! Make learning fun!

Watch for our next computer games guidebook, Values, Creativity, Resources, and the Computer which focuses on resource management, value judgement, and creative thinking. Also watch for our forthcoming booklet, How to Write a Computer Simulation.

> "To be practical, an education should prepare a man for work that doesn't yet exist, and whose nature cannot even be imagined."
>
> Charles E. Silberman

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